Project Catalogue

Modeling and Managing Electricity Risks

Instructions for all projects

- The goal is to do some data analysis on the indicated topic. References are provided for background and inspiration, you don't have to follow the methodology of the paper. Be creative in the choice additional data sources and additional references, which should be quoted in your report.
- The deliverable is a pdf report (5-8 pages) + zip file with python code and all data files required to run the code. You can also put everything (report + code) into a Jupyter notebook.
- The reports will be graded according to the following points:
  - Understanding of the energy risk management problem
  - Clarity of model description
  - Interpretation of the results
  - Quality of visualizations
- Main data sources:
  - Renewables Ninja `https://www.renewables.ninja/` for wind and solar power production and weather data. Create a free account to dowload large quantities of data.
  - ENTSO-E `https://transparency.entsoe.eu/` for information related to load, generation and market data.
  - SIRTA `https://sirta.ipsl.fr/` for high frequency weather data at IP Paris site.

  Non-public data will be provided for specific projects. These should be treated as confidential and erased at the end of the project.

Contents

1 Stochastic Modeling and Risk Analysis of Day-Ahead Electricity Prices

Context

Electricity spot prices exhibit strong seasonality, spikes, mean-reversion and regime changes. The goal of this project is to build and compare simple stochastic models for day-ahead electricity prices in a chosen European market, and to assess their implications for risk management.

Objectives

- Explore the statistical properties of day-ahead prices (seasonality, spikes, autocorrelation).
- Fit at least two stochastic models (e.g. those seen in the lectures, mean-reverting with jumps, regime-switching etc.).
- Simulate price paths and compute risk measures for different time horizons.

Data

Hourly or daily day-ahead prices for 3–5 years from one country (e.g. ENTSO-E). Optionally, load and fuel prices can be included for interpretation.

Suggested Tasks

1. Exploratory analysis: plots, histograms, seasonal decomposition, autocorrelation.
2. Detrend / deseasonalise the series and fit simple models:
3. Estimate parameters (e.g. via maximum likelihood or least squares).
4. Simulate multiple price paths and:
    - Compute distribution of average price over different horizons,
    - Compute Value-at-Risk / Expected Shortfall of price over 1 day, 1 week, 1 month.
5. Compare the models in terms of statistical fit and risk estimates.

Reference: Aïd, R., Campi, L., Huu, A. N., & Touzi, N. (2009). A structural risk- neutral model of electricity prices. International Journal of Theoretical and Applied Finance, 12(07), 925-947.

2 Quantifying Imbalance Risk for a Wind Portfolio in Day-Ahead and Balancing Markets

Context

Wind producers commit to day-ahead schedules but actual generation is uncertain. Deviations are settled at imbalance or balancing prices, creating imbalance cost and risk.

Objectives

- Quantify imbalance volumes and imbalance costs for a stylised wind portfolio.
- Explore how forecast errors translate into financial risk.
- Compare alternative scheduling strategies.

Data

- Aggregated actual wind generation (Renewables Ninja),
- Day-ahead prices (ENTSO-E) and imbalance (balancing) prices for the same period (`https://www.services-rte.com/en/view-data-published-by-rte/balancing.html`), create a free account to download data).

Suggested Tasks

1. Construct a stylised wind portfolio (by aggregating several existing or hypothetical stations).
2. Build a simple wind forecast (e.g. based on historical average, persistence, or basic time series model).
3. Define day-ahead schedule as the forecast and compute imbalance:

$$\text{Imbalance}_t = \text{Actual}_t - \text{Schedule}_t.$$

4. Using imbalance pricing rules, compute imbalance costs and total revenue (spot + imbalance settlement).
5. Compare:
    - Naive schedule (e.g. persistence),
    - Improved forecast-based schedule,
    - Conservative schedule (e.g. lower quantile of forecast).
6. Analyse the distribution of profit, imbalance volumes and associated risk measures.

Reference: Dupré, A., Drobinski, P., Badosa, J., Briard, C., & Tankov, P. (2020). The economic value of wind energy nowcasting. Energies, 13(20), 5266.

3 Hydro Reservoir Management Under Price Uncertainty: Dynamic Programming vs. Heuristics

Context

Hydro reservoirs provide flexibility by shifting energy across time. Optimal management under uncertain prices is a classic stochastic control problem, often solved by dynamic programming or simple heuristics.

Objectives

- Formulate a simplified hydro scheduling problem under uncertain prices.
- Implement a dynamic programming (DP) solution for a small state space.
- Compare DP with heuristic strategies (e.g. fixed rule or policy based on price thresholds).

Data

Hourly or daily day-ahead prices over several years. Optionally, inflow data can be stylised or simplified (e.g. constant inflow).

Suggested Tasks

1. Define a simplified hydro plant:
   - Reservoir capacity, initial level, inflows,
   - Maximum turbine power, spill option.
2. Model prices either:
   - Directly from historical time series in a rolling-horizon setting, or
   - Via a simple stochastic model (e.g. 2-state Markov chain for high/low prices).
3. Implement a DP algorithm (finite horizon, discrete state for reservoir level) to maximise expected revenue.
4. Implement one or two heuristic policies (e.g. "produce if price exceeds threshold, otherwise store").
5. Compare the performance of DP and heuristics in terms of expected revenue and variability over simulated price paths.

Reference: Steffen, B., & Weber, C. (2016). Optimal operation of pumped-hydro storage plants with continuous time-varying power prices. European Journal of Operational Research, 252(1), 308-321.

4 Design and Risk Evaluation of Time-of-Use and Dynamic Retail Tariffs

Context

Retail electricity tariffs can be flat, time-of-use (TOU) or dynamic (indexed to wholesale prices). These choices affect both consumers and suppliers in terms of cost, risk and load profiles.

Objectives

- Design simple TOU and dynamic tariffs based on wholesale prices.
- Evaluate the impact on retailer margin and risk, given a load profile.
- Explore how consumer flexibility could change results.

Data

- Wholesale day-ahead prices over several years,
- Typical household or small business load profiles (see e.g., `https://figshare.com/articles/dataset/ELMAS_dataset/23889780`)

Suggested Tasks

1. Define a base flat tariff and simple TOU tariff(s) (e.g. peak/off-peak rates).
2. Define a dynamic tariff indexed to wholesale prices (e.g. markup over day-ahead price).
3. For a given load profile, compute:
   - Consumer annual cost under each tariff,
   - Retailer margin = tariff revenue − wholesale cost.
4. Compute risk measures for the retailer margin (variance, VaR) under each tariff type.
5. Introduce a simple consumer response model (e.g. ability to shift a fraction of load from peak to off-peak) and analyse how this changes results.

5 Valuing Demand Response as a Flexibility Resource in Spot and Balancing Markets

Context

Demand response (DR) can provide flexibility by reducing or shifting consumption in response to price signals. This project evaluates the economic value and risk profile of a DR portfolio.

Objectives

- Model a simple DR product that responds to high prices or explicit activation.
- Quantify revenues from DR in spot and/or balancing markets.
- Analyse risk and compare with alternative flexibility options.

Data

- Historical spot prices (and optionally balancing prices),
- Typical load profiles for participating consumers (see e.g., `https://figshare.com/articles/dataset/ELMAS_dataset/23889780`)

Suggested Tasks

1. Specify a DR scheme (e.g. load sheds when price exceeds a threshold or upon operator request).
2. Define technical limits: maximum reduction, number of activations per day/week, rebound effect.
3. Compute DR revenues under a given remuneration scheme (e.g. capacity payment + activation payment).
4. Evaluate the impact on consumer cost and on the aggregator's revenue and risk.
5. Explore different thresholds, capacities and price scenarios.

6 Modeling Congestion and Pricing Transmission Rights in Coupled Power Markets

Context

In coupled power markets, congestion between bidding zones leads to price differences and congestion rents. Financial transmission rights (FTRs) or similar instruments can hedge these price spreads.

Objectives

- Analyse historical price differences between two coupled bidding zones.
- Model congestion events and their frequency/intensity.
- Evaluate the value and risk of a simple transmission right (spread option) between the two zones.

Data

- Hourly day-ahead prices for two interconnected bidding zones (e.g. FR-DE, FR-ES)

Suggested Tasks

1. Compute and analyse hourly price spreads between the two zones.
2. Identify congestion patterns (frequency, duration, magnitude).
3. Define a simple FTR-like product: right to receive the average spread over a month or quarter.
4. Price this product using:
    - Historical average of spreads,
    - Simple stochastic model for spreads (e.g. two-state Markov chain: uncongested/congested).
5. Compute risk measures on the value of the FTR from the holder's perspective.

7 Co-Optimization of Renewable Generation and Battery Storage for Producer Revenues

Context

Combining variable renewables (wind/solar) with battery storage can improve producer revenue by smoothing output and shifting energy to high-price hours. This project studies co-optimisation of a renewable plant plus battery.

Objectives

- Model joint operation of a renewable plant and a battery in the day-ahead market.
- Optimise dispatch to maximise expected revenue under price uncertainty.
- Compare risk/return with and without storage.

Data

- Hourly renewable generation profile (wind or solar) for several years (Renewables Ninja),
- Hourly day-ahead prices for the same period (ENTSO-E).

Suggested Tasks

1. Define a stylised wind or solar plant (capacity scaling of actual data).
2. Introduce a battery model (energy capacity, power limits, efficiency).
3. Formulate an optimisation problem (e.g. daily or weekly) to:
   - Decide how much renewable energy to sell immediately vs. store,
   - Decide charge/discharge schedule of the battery.
4. Consider:
   - Perfect foresight optimisation,
   - Simple forecast-based rolling-horizon strategy.
5. Compare:
   - Mean annual revenue and volatility without storage,
   - Mean annual revenue and volatility with storage.
6. Discuss how storage size and efficiency influence the risk/return profile.

Reference: H. Ding, P. Pinson, Z. Hu, and Y. Song, Integrated bidding and operating strategies for wind-storage systems, IEEE Transactions on Sustainable Energy, 7 (2016), pp. 163–172.

8 Load Modeling and Forecasting (Short-Term)

Context

Short-term load forecasting (from a few hours to a few days ahead) is a key input for scheduling generation, managing reserves and bidding in day-ahead markets. Load depends on calendar effects, temperature and other factors, and exhibits strong intraday and weekly patterns.

Objectives

- Analyse the statistical properties of hourly electricity demand.
- Build and compare short-term load forecasting models (e.g. persistence, linear regression, ARIMA).
- Evaluate forecasts with appropriate error metrics and discuss sources of error.

Data

- Hourly system load / demand for one country or control area (3–5 years) (ENTSO-E).
- Hourly temperature data for one or more representative weather stations (e.g., Meteo France open data `https://www.data.gouv.fr/datasets/donnees-climatologiques-de-base-quotidiennes/` or other public data sources).
- Calendar information (day-of-week, holidays).

Suggested Tasks

1. Perform exploratory analysis:
   - Plot daily and weekly load profiles,
   - Decompose seasonality (daily / weekly / annual),
   - Examine relationship between load and temperature.
2. Construct and compare at least three forecasting models:
   - Benchmark: persistence or simple average of similar past days,
   - Linear regression with temperature and calendar features,
   - ARIMA/SARIMA or another time series model (optionally with exogenous inputs).
3. Design a rolling-origin evaluation (e.g. moving training window):
   - Forecast 24 hours ahead each day for a test period,
   - Compute MAE, RMSE, MAPE, and compare models.
4. Analyse forecast errors by hour of day and by season, and discuss practical implications for system operators and market participants.

9 Winter Load and Wind Profiles and Required Generation Margin

Context

System operators and planners need to ensure that there is sufficient firm generation capacity to cover demand during the winter season, when loads are typically highest. At the same time, a growing share of wind power means that the *net* load (load minus wind generation) can be very different from the raw load. This project focuses on understanding the distribution of hourly load on a *typical winter day* in the coming season, comparing it with typical wind generation, and using this information to infer the required generation margin.

Objectives

- Characterise the distribution of hourly load on winter days (e.g. December–February) using recent historical data.
- Characterise the distribution of hourly wind generation over the same period.
- Construct scenarios for a "typical" winter day in the coming season for both load and wind.
- Derive the distribution of *net load* (load minus wind) and estimate the firm generation margin required to achieve a given reliability level.

Data

- Historical hourly system load (demand) for a chosen country or control area for several past winters (e.g. last 5–10 years).
- Historical hourly wind generation (aggregated) for the same area and period.
- Optional: temperature data to help interpret inter-winter differences (cold vs. mild winters).

Suggested Tasks

1. Select winter days (e.g. December–February) for the last 5–10 years and clean the load and wind data.
2. For each hour of the day, compute and plot typical winter profiles (mean/median and a few quantiles) for load and wind; highlight morning and evening peaks, and low-wind situations.
3. Build a simple "coming winter" scenario by adjusting historical profiles for recent trends (e.g. modest load growth or wind capacity increase).
4. Compute net load $N_t = L_t - W_t$ and estimate the distribution of its maximum on a winter day or over the whole winter.
5. For a given reliability level (e.g. 99% of winter hours served), estimate the firm capacity margin (in MW and as % of peak load) that would be required.

10 Probabilistic Forecasting of Wind Resource

Context

Wind power is highly uncertain. Probabilistic forecasts (full predictive distributions or quantiles) are more informative than point forecasts and are essential for risk-aware decision-making (e.g. bidding, reserve sizing).

Objectives

- Build probabilistic forecasts for wind speed.
- Compare different methods (e.g. parametric distributions, quantile regression).
- Evaluate forecast quality using proper scoring rules.

Data

- Time series of hourly wind speed at one or more locations.

Suggested Tasks

1. Exploratory analysis of wind data:
    - Seasonality and diurnal patterns,
    - Empirical distribution (skewness, heavy tails).
2. Choose and implement at least two probabilistic forecasting methods, for example:
    - Parametric model (e.g. truncated normal or Weibull) with parameters depending on predictors,
    - Quantile regression (linear or non-linear) for several quantiles,
3. Evaluate forecasts using:
    - CRPS (continuous ranked probability score),
    - Pinball loss for selected quantiles,
    - Reliability diagrams and coverage of prediction intervals.
4. Compare methods and discuss strengths/weaknesses from the perspective of a wind producer or system operator.

Reference: Gneiting, Tilmann, and Matthias Katzfuss, Probabilistic forecasting. Annual Review of Statistics and Its Application 1 (2014): 125-151.

11 Probabilistic Forecasting of Solar Resource

Context

Solar power output depends strongly on cloud cover and exhibits strong diurnal and seasonal patterns. Probabilistic solar forecasts help in managing reserve needs, storage and bidding strategies.

Objectives

- Develop probabilistic forecasts for solar irradiance.
- Account for daily and seasonal patterns and weather variability.
- Evaluate probabilistic forecasts using suitable metrics.

Data

- Time series of solar irradiance.
- Solar geometry information (clear-sky irradiance).

Suggested Tasks

1. Transform raw data to a normalised measure (e.g. clear-sky index: actual / clear-sky).
2. Build probabilistic models such as:
   - Parametric distributions for the normalised output (e.g. Beta distribution),
   - Quantile regression as a function of lagged values, time of day, and day of year.
3. Evaluate:
   - Sharpness and calibration of prediction intervals,
   - CRPS and pinball loss on a test set,
   - Differences across seasons or weather regimes.
4. Discuss how such forecasts could be used by a PV operator in market bidding or storage operation.

Reference: Badosa, J., Gobet, E., Grangereau, M., & Kim, D. (2017). Day-ahead probabilistic forecast of solar irradiance: a Stochastic Differential Equation approach. `https://hal.archives-ouvertes.fr/hal-01625651/`

12 Modeling of Wind Power Curve

Context

The wind power curve links wind speed to turbine or farm power output. In practice, power curves differ from theoretical ones due to turbulence, wakes, measurement errors and curtailments.

Objectives

- Empirically estimate a power curve from data (wind speed, power output).
- Compare different functional forms or non-parametric approaches.
- Analyse residuals and discuss sources of deviation from the ideal power curve.

Data

- Paired time series of wind speed (at hub height or corrected) and power output for a turbine or wind farm (SCADA data if available, or aggregated data as a proxy). See e.g., `https://www.kaggle.com/datasets/mubashirrahim/wind-power-generation-data-forecasting`

Suggested Tasks

1. Clean the data:
    - Remove obvious outliers, maintenance periods, curtailment if identifiable.
2. Fit at least two power curve models:
    - Parametric model (e.g. piecewise polynomial or logistic curve with cut-in, rated and cut-out speeds),
    - Non-parametric model (e.g. local regression, spline, or bin-averaged curve).
3. Compare models via:
    - RMSE or MAE on a validation set,
    - Visual inspection of fitted vs. observed curves,
    - Analysis of residuals vs. wind speed.
4. Investigate how environmental factors (e.g. turbulence intensity, direction, time of day) may affect the power curve if such data are available.

Reference: Fischer, A., Montuelle, L., Mougeot, M., & Picard, D. Statistical learning for wind power: A modeling and stability study towards forecasting. Wind Energy.

13 Estimating Profitability of a Wind Park

Context

Investment in a wind park requires evaluating expected revenues, costs and risk over the project lifetime. This project focuses on building a simple financial model using historical wind and price data.

Objectives

- Construct a cash-flow model for a wind park (CAPEX, OPEX, revenues).
- Estimate profitability metrics (NPV, IRR, payback time).
- Assess the impact of price and resource uncertainty on profitability.

Data

- Historical hourly wind generation profile (Renewables Ninja),
- Historical hourly spot prices,
- Stylised investment parameters: CAPEX (€/MW), fixed and variable OPEX, lifetime, discount rate.

Suggested Tasks

1. Define the characteristics of a hypothetical wind park (capacity, location, lifetime).
2. Compute yearly energy production using the wind profile.
3. Under a given subsidy scheme (exposed to spot prices), compute yearly revenues and a discounted cash-flow over the lifetime.
4. Calculate:
    - NPV, IRR, simple payback time,
    - Sensitivity to CAPEX, OPEX, and discount rate.
5. Explore alternative revenue schemes (e.g. feed-in tariffs or feed-in premiums) and compare risk/return profiles.
6. Use bootstrap or simple stochastic price/wind models to assess the distribution of NPV (profitability risk).

14 Estimating Profitability of a Solar Park

Context

Solar PV investments face similar issues as wind, but with different generation profiles, correlation to prices and technology costs. This project focuses on evaluating the profitability of a solar park.

Objectives

- Build a cash-flow model for a PV park.
- Estimate NPV and IRR under different support schemes.
- Analyse how irradiance variability and price patterns affect profitability.

Data

- Historical hourly or sub-hourly PV production data or irradiance data converted via a simple PV model (Renewables Ninja),
- Historical spot prices,
- Assumed CAPEX and OPEX for PV (€/kW, €/kW/year).

Suggested Tasks

1. Define a PV park (capacity, location, tilt/orientation assumptions).
2. Build a production time series from data or via a simple PV performance model.
3. Compute revenues under:
   - Direct exposure to spot prices,
   - Feed-in tariff or premium-type support schemes.
4. Calculate NPV, IRR and payback and perform sensitivity analysis to:
   - CAPEX, OPEX,
   - Price level and volatility,
   - Degradation rate of panels.

15 Geographical Correlation of Wind Resource and Wind Power Production

Context

Spatial diversification of wind sites can reduce aggregate variability and forecast errors. Understanding geographical correlations is crucial for portfolio design, transmission planning and system operation.

Objectives

- Quantify spatial correlations of wind speeds or wind power outputs across multiple locations.
- Study how aggregation reduces variability and ramping.
- Discuss implications for portfolio design and system integration of wind.

Data

- Time series of wind power or wind speed from several geographically distinct locations (or regions),
- Approximate locations (coordinates or region identifiers).

Suggested Tasks

1. Compute pairwise correlation coefficients of wind power / wind speed between sites.
2. Analyse how correlation depends on distance and direction between sites.
3. Construct aggregated portfolios (e.g. equal weights, regional mixes) and compare:
    - Variance of total output,
    - Frequency of large ramps (sudden changes),
    - Duration of low or high production events.
4. Discuss implications for transmission needs and the value of geographical diversification.

16 Post-Processing of Ensemble Forecasts

Context

Ensemble forecasts (e.g. from NWP models) provide multiple scenarios for weather variables or renewable generation. Raw ensembles are often biased and under-dispersed and need statistical post-processing to improve reliability.

Objectives

- Apply statistical post-processing methods to an ensemble forecast (for wind, solar, or load).
- Improve calibration and sharpness of predictive distributions.
- Evaluate performance using probabilistic verification tools.

Data

- Historical ensemble forecasts will be provided to the students on request (to be erased after end of project)
- Corresponding observations for the same locations and times (SIRTA).

Suggested Tasks

1. Evaluate raw ensemble:
   - Rank histogram,
   - CRPS and other scores.
2. Implement at least one post-processing method, such as:
   - EMOS (ensemble model output statistics)
3. Compare raw vs post-processed ensemble in terms of:
   - Bias, dispersion, reliability,
   - CRPS score for events.
4. Discuss the impact of improved probabilistic forecasts on operational decisions (e.g. reserve sizing, hedging).

References: Gneiting, T., Raftery, A. E., Westveld III, A. H., & Goldman, T. (2005). Calibrated probabilistic forecasting using ensemble model output statistics and minimum CRPS estimation. Monthly weather review, 133(5), 1098-1118.

17 Using Battery Storage for Day-Ahead Arbitrage (price taker)

Context

A battery can earn revenue by buying energy in low-price hours and selling in high-price hours in the day-ahead market. This project is a focused study of day-ahead arbitrage strategies using a battery model. In this project we take the day-ahead prices as given.

Objectives

- Model a battery and its operational constraints.
- Formulate a day-ahead arbitrage optimisation problem.
- Evaluate revenues and simple risk indicators under historical prices.

Data

- Historical hourly day-ahead prices in a chosen market.

Suggested Tasks

1. Define a battery (energy capacity, power, efficiency, initial state of charge).
2. For each day, solve a deterministic optimisation problem:
    - Maximise daily arbitrage profit given known day-ahead prices subject to battery constraints.
3. Aggregate results over the full data set:
    - Compute annual revenues, utilisation hours, average cycles.
4. Analyse the distribution of daily profits and simple risk measures (e.g. probability of negative profit days).
5. Perform sensitivity analysis with respect to battery size and efficiency.

18 Using Battery Storage for Day-Ahead Arbitrage (price maker)

Context

A battery can earn revenue by buying energy in low-price hours and selling in high-price hours in the day-ahead market. This project is a focused study of day-ahead arbitrage strategies using a battery model. In this project we take bid-offer curves as given, and assume that the bid/offer of the battery is added to the corresponding curve, which then modifies the price calculation.

Objectives

- Model a battery and its operational constraints.
- Formulate a day-ahead arbitrage optimisation problem.
- Evaluate revenues and simple risk indicators under historical bid/offer curve data.

Data

- Bid and offer curves will be provided to the students on request (to be erased after end of project)

Suggested Tasks

1. Define a battery (energy capacity, power, efficiency, initial state of charge).
2. For each day, solve a deterministic optimisation problem:
    - Maximise daily arbitrage profit given known bid/offer curves subject to battery constraints.
3. Aggregate results over the full data set:
    - Compute annual revenues, utilisation hours, average cycles.
4. Analyse the distribution of daily profits and simple risk measures (e.g. probability of negative profit days).
5. Perform sensitivity analysis with respect to battery size and efficiency.

19 Seasonal Optimal Mix of Wind and Solar in France

Context

The optimal mix of wind and solar capacities depends on their seasonal profiles, correlation with demand and prices, and risk considerations. This project explores the seasonal complementarity of wind and solar in France and identifies an optimal portfolio mix.

Objectives

- Analyse seasonal profiles of wind and solar generation and their correlation with load and prices.
- Formulate an optimisation problem to choose wind/solar capacity shares.
- Evaluate risk/return trade-offs for different mixes.

Data

- Historical time series of wind and solar generation (or capacity factors) in France,
- Historical load (demand) and spot prices for the same period.

Suggested Tasks

1. Compute seasonal (monthly or quarterly) averages and variability of:
   - Wind capacity factor,
   - Solar capacity factor,
   - Load and prices.
2. Examine correlation patterns:
   - Wind vs. solar,
   - Each technology vs. load and prices.
3. Consider a portfolio with capacities C_{wind} and C_{solar}:
   - Compute revenue time series for different mixes (e.g. fixed total capacity and varying shares).
4. Evaluate for each mix:
   - Mean annual revenue,
   - Revenue volatility and downside risk.
5. Identify mixes that optimise:
   - Expected revenue,
   - Or a mean-variance trade-off (efficient frontier).
6. Discuss seasonal complementarity and implications for planning and policy.

Reference: Heide, Dominik, et al. "Seasonal optimal mix of wind and solar power in a future, highly renewable Europe." Renewable Energy 35.11 (2010): 2483-2489.

20 Modeling and Hedging Clean Spark Spread Risk for a Gas-Fired Power Plant

Context

A gas-fired power plant's margin depends on the *clean spark spread* (CSS), i.e. electricity price minus fuel and CO_2 costs. CSS is volatile and exposed to multiple markets (power, gas, carbon). This project focuses on modeling CSS and performing various risk estimates.

Objectives

- Construct and analyse a historical time series of clean spark spreads.
- Model CSS dynamics using simple stochastic models.
- Evaluate various risk indicators.

Data

- Historical spot (or day-ahead) power prices,
- Historical gas hub prices (`https://tradingeconomics.com/commodity/eu-natural-gas`)
- Historical CO_2 (EUA) prices (`https://www.investing.com/commodities/carbon-emissions-historica`
- Assumed plant heat rate and emission factor.

Suggested Tasks

1. Construct a clean spark spread series:

$$\mathrm{CSS}_t = P_t^{\mathrm{power}} - \alpha P_t^{\mathrm{gas}} - \beta P_t^{\mathrm{CO_2}},$$

   where α is the heat rate and β the emission factor.

2. Perform exploratory analysis:
   - Distribution, autocorrelation, seasonality of CSS,
   - Relation to underlying prices.
3. Model CSS using simple approaches (e.g. AR(1), mean-reverting model) and estimate parameters.
4. Specify a stylised plant (capacity, fixed O&M, start-up constraints may be ignored for simplicity) and compute revenue distribution over historical data. Evaluate mean and variance of profits and various risk measures (VaR, expected shortfall).

21 Modeling Negative Price Events and Curtailment Risk in High-Renewables Power Systems

Context

In systems with high shares of renewables, spot prices may become very low or negative during periods of oversupply, leading to curtailment and revenue risk for generators. Understanding the frequency and drivers of such events is essential.

Objectives

- Analyse the occurrence and characteristics of negative (or very low) price events.
- Identify key explanatory variables (e.g. wind, solar, load, interconnector flows).
- Model the probability of negative prices and discuss implications for curtailment risk.

Data

- Historical hourly spot prices for a market with occasional negative prices (e.g., German market),
- Corresponding hourly data for load, wind and solar generation, and possibly cross-border flows.

Suggested Tasks

1. Identify and characterise negative price events (duration, timing, seasonality).
2. Explore relationships between negative prices and:
   - High renewable output,
   - Low demand,
   - Congestion or export constraints.
3. Specify and estimate a simple classification model for negative price occurrence, e.g. logistic regression:
   $$\mathbb{P}(P_t < 0 \mid X_t) = f(X_t),$$
   where X_t includes load, wind, solar, etc.
4. Evaluate model performance.
5. Discuss how the frequency and predictability of negative prices affect:
   - Revenue risk for renewable generators,
   - The need for curtailment and flexibility resources.

22 Reserve Dimensioning Using Probabilistic Load and Renewable Generation Forecasts

Context

System operators must determine reserve levels to guard against forecast errors in load and renewable generation. Probabilistic forecasts can support risk-based reserve sizing.

Objectives

- Combine probabilistic forecasts of load and renewables to assess net load uncertainty.
- Propose a rule for reserve dimensioning based on a reliability criterion.
- Evaluate trade-offs between reliability (shortage probability) and reserve cost.

Data

- Historical time series of load and renewable (wind/solar) generation,
- Assumed cost of holding reserves (€/MW) (see `https://www.services-rte.com/en/view-data-publishe` `balancing-capacities.html` for actual reserve prices).

Suggested Tasks

1. Develop a simple model for probabilistic day-ahead forecasting of *net load* (load minus renewables).
2. Define a reserve requirement R_t such that:

$$\mathbb{P}(\text{Net load error} > R_t) \leq \varepsilon,$$

   for a target risk level ε (e.g. 1%).

3. Simulate reserve requirements over a test period and compute:
   - Average reserve level,
   - Frequency of under-coverage (events where realised error exceeds R_t),
   - Total reserve cost.
4. Compare alternative policies (different ε, conservative margins, etc.) in terms of cost and reliability.

23 Co-Optimization of Battery Storage Between Energy Arbitrage and Frequency Reserve Markets

Context

Batteries can provide both energy arbitrage (buy low, sell high) and frequency reserves. Co-optimizing participation in these markets can significantly affect revenues and risk.

Objectives

- Model a battery capable of both energy arbitrage and reserve provision.
- Formulate a co-optimization problem deciding how much capacity to allocate to each service.
- Evaluate revenue and risk under historical price and reserve remuneration data.

Data

- Historical hourly day-ahead prices,
- Historical reserve prices or capacity payments for frequency regulation / primary or secondary reserve (see `https://www.services-rte.com/en/view-data-published-by-rte/balancing-capacities.html`),
- Battery technical parameters (capacity, power, efficiency).

Suggested Tasks

1. Specify the battery model with constraints on energy and power.
2. Define decision variables:
    - Hourly charging/discharging for arbitrage,
    - Reserve capacity allocated in each hour (or block).
3. Formulate an optimisation problem over a representative horizon:
    - Maximise total revenue from arbitrage + reserve capacity payments,
    - Subject to battery constraints and possibly minimum state-of-charge for reserve provision.
4. Implement and solve the problem using `cvxpy` or similar.
5. Compare:
    - Pure arbitrage vs pure reserve vs co-optimised solution,
    - Revenues, utilisation and risk characteristics.

24 Impact of Electric Vehicle Charging on Distribution-Level Load Peaks and Risk

Context

The electrification of transport increases electricity demand, especially at the distribution level. EV charging profiles can create new peaks and increase network stress, unless managed smartly.

Objectives

- Model EV charging demand for a given population of vehicles.
- Assess the impact on distribution feeder load profiles and peak demand.
- Compare uncontrolled charging with simple smart charging strategies.

Data

- Baseline distribution-level load profiles (see e.g., `https://figshare.com/articles/dataset/ELMAS_dataset/23889780`)
- Assumed EV usage and charging patterns (arrival times, energy needs),
- Charging power and battery sizes typical for EVs.

Suggested Tasks

1. Generate synthetic EV charging demand based on assumptions about:
   - Number of EVs on the feeder,
   - Arrival/departure times,
   - Daily driving distances and required energy.
2. Superimpose EV demand on baseline load to obtain total load profiles.
3. Compute and compare:
   - Peak demand levels,
   - Load factor and variability,
   - Frequency of hours above certain thresholds.
4. Design simple smart charging schemes (e.g. delayed charging, valley filling) and simulate their impact.
5. Discuss implications for network reinforcement, tariffs and incentives for smart charging.

25 Risk and Valuation of Capacity Mechanisms (Capacity Markets and Reliability Options)

Context

In many liberalised power systems, an *energy-only* market may not provide sufficient incentives to invest in firm capacity, especially when scarcity prices are politically constrained. Capacity mechanisms (e.g. capacity markets, reliability options) aim to ensure adequacy by paying generators (and demand response) for available capacity in addition to energy. This project studies how such mechanisms affect the risk and expected value of revenues for a generator, and the cost and reliability for consumers.

Objectives

- Understand the basic design of a capacity mechanism (simple capacity payments or reliability options).
- Build a cash-flow model for a representative generator with and without a capacity mechanism.
- Quantify changes in expected revenue, revenue risk and adequacy (e.g. loss-of-load risk) under different designs.

Data

- Historical hourly spot prices and load for a chosen market (several years).

Suggested Tasks

1. **Baseline energy-only model**
   - Specify a representative thermal generator (capacity K, variable cost c^{var}, fixed cost c^{fix}).
   - Using historical spot prices and assumed dispatch rule (e.g. plant runs when price $> c^{\text{var}}$), compute:
     - Annual energy produced and revenues,
     - Operating profit (energy margin minus fixed cost) and its variability.
2. **Capacity market design**
   - Assume a capacity payment p^{cap} (€/MW/year) for committed capacity K.
   - Add this to the generator's cash-flow and recompute:
     - Expected annual profit,
     - Standard deviation / downside risk (e.g. Value-at-Risk).
   - Explore scenarios for different levels of p^{cap} and discuss under which conditions the plant becomes financially viable.
3. **Reliability options design**
   - Model a reliability option contract: the generator receives a premium (capacity payment) but must pay back $(P_t - K_{\text{strike}})^+$ when spot price P_t exceeds the strike K_{strike}.
   - Compute net revenue time series:
     - Energy revenues (as before),

  - Premium (capacity payment),
  - Option paybacks during scarcity hours.
- Compare risk and expected revenue under:
  - Energy-only,
  - Pure capacity payment,
  - Reliability option.

26 Extreme Temperature Events, Peak Demand and Loss-of-Load Risk Analysis

Context

Extreme temperature events (cold waves or heat waves) can drive electricity demand to very high levels (heating or cooling), increasing the risk of supply shortages. Understanding the relationship between temperature extremes, peak demand and loss-of-load risk is crucial for adequacy planning and for assessing the benefits of demand response, storage or interconnections.

Objectives

- Quantify the relationship between temperature and peak electricity demand.
- Analyse how extreme temperature events translate into extreme load events.
- Estimate loss-of-load risk for a given installed firm capacity under historical conditions and stylised future scenarios.

Data

- Historical hourly (or daily) electricity demand for a country/region over several years.
- Corresponding temperature data (e.g., Meteo France open data `https://www.data.gouv.fr/datasets/donnees-climatologiques-de-base-quotidiennes/` or other public data sources).

Suggested Tasks

1. **Exploratory analysis**
   - Plot load vs. temperature scatterplots for winter and summer separately.
   - Identify approximate temperature sensitivity (e.g. piecewise linear response for heating and cooling).
   - Highlight historical extreme temperature events and the corresponding load peaks.
2. **Temperature–load model**
   - Estimate a simple regression model of peak (or daily maximum) load as a function of temperature and calendar variables (day-of-week, season).
3. **Loss-of-load risk for given firm capacity**
   - For a chosen firm capacity level C^{firm}, compute:
     - The number of hours in each year where load exceeds C^{firm} (historical *loss-of-load hours*),
     - Statistics such as Loss-of-Load Expectation (LOLE; expected number of hours per year with $L_t > C^{\text{firm}}$).
   - Explore how LOLE changes as a function of C^{firm} (e.g. plot LOLE vs. capacity).
4. **Scenario and sensitivity analysis**
   - Construct simple scenarios for increased temperature volatility or more frequent extremes (e.g. shift in temperature distribution based on climate projections).
   - Using the temperature–load model, simulate how these scenarios would affect peak load and LOLE.

27 Modeling and Pricing Structured Supply Contracts (Swing or Take-or-Pay) in Power Markets

Context

Structured contracts such as swing options give the buyer flexibility in the volume of energy taken, often subject to constraints.

Objectives

- Specify a simple structured supply contract (e.g. monthly swing, daily volume flexibility).
- Model underlying spot or prices.
- Estimate the contract value and risk using simulation.

Data

- Historical spot prices.

Suggested Tasks

1. Define the contract payoff structure, including:
   - Daily or hourly decision on how much energy to take,
   - Global constraints (e.g. annual min/max volumes).
2. Choose one or several price models (e.g. log-normal mean-reverting or jump process).
3. Implement a simulation-based valuation:
   - Determine the optimal exercise decisions for the buyer (possibly via dynamic programing or some ad hoc strategies).
   - Estimate the expected contract value and its distribution.
4. Analyse sensitivity of contract value to:
   - Strike price and flexibility limits,
   - Price volatility.

Reference: Jaillet, P., Ronn, E. I., & Tompaidis, S. (2004). Valuation of commodity-based swing options. Management science, 50(7), 909-921.